

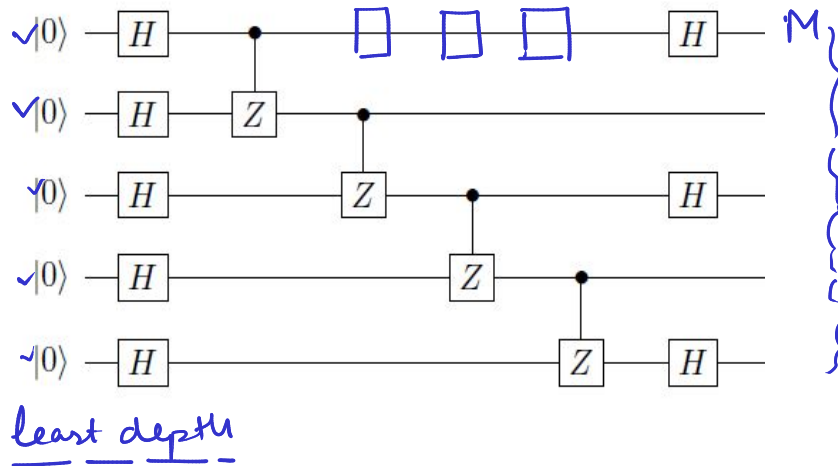
The quantum computational model

Classical

- Representation of information
 - Bit: fundamental unit of classical information
 $\{0,1\}$
- Computation (state-changes) *$f: \{0,1\}^n \rightarrow \{0,1\}^m$*
 - Modeled by Boolean functions
 - Implemented by digital logic circuits
 - Universal gate-set (AND, NOT) *↖*
 - Reversible with extra (redundant) bits

Quantum

- Representation of information
 - Qubit: fundamental unit of quantum information
- Computation (state-changes) *qubits*
 - Modeled by unitary evolution *U*
 - Implemented by quantum logic circuits
 - Universal gate-set (U3, CNOT) ✓
 - Fundamentally reversible (unitary evolution) *$U \rightarrow \text{rev.}$
 $M \rightarrow \text{inv.}$*
 - Quantum algorithm
 - Superposition, entanglement, interference



- Complexity Measures
 - ← "intuition" → Computational complexity → #steps
 - ✓ Query complexity → *Grover's*
 - ✓ Communication complexity

- Complexity Classes
 - BQP ✓ ⇒
 - QMA ✓

From classical to quantum computation

representation ✓
computation ✓

(TM) $\Rightarrow$ undecidable $\rightarrow$

• Computability and Complexity

- The notion of computability does not change
 - $\rightarrow$ Definition of effective computation is same for quantum and classical models
- The notion of complexity does change
 - $\rightarrow$ Computation in Hilbert-space : exponential advantage in efficiency

(NP)
 $\downarrow$
polynomial

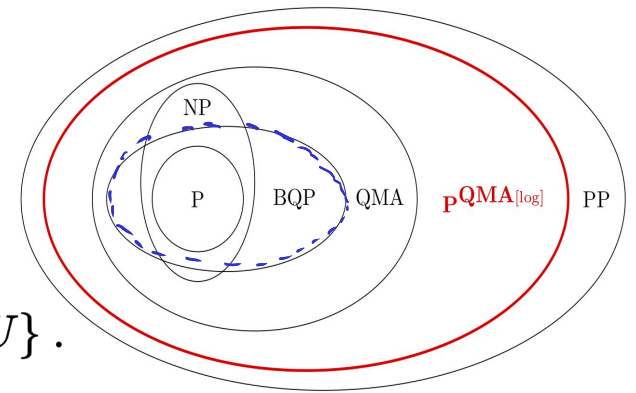
• Quantum Advantage

- Efficient quantum algorithms have been discovered for specific problems
 - $\rightarrow$ which are **easy (polynomially-solvable)** for quantum computers
 - $\rightarrow$ which are **believed to be hard** for classical computers
 - $\rightarrow$ Example: Shor's quantum algorithm for integer factorization

$\rightarrow$ FACTORING $:= \{ \langle N, L, U \rangle : \exists \text{ prime } p \text{ s.t. } p \mid N, L \leq p \leq U \}$.

✓ FACTORING-SEARCH: $p_1 \cdot \dots \cdot p_n = N$

$$\exp \left(\left(\sqrt[3]{\frac{64}{9}} + o(1) \right) (\ln n)^{\frac{1}{3}} (\ln \ln n)^{\frac{2}{3}} \right) \quad \text{classical} \quad O(n^2 \log^d(n/\varepsilon)) \quad \text{quantum} \quad \Rightarrow \underline{O(n^3)}$$



efficiency $[f(n)]^{2/3} =$
 $\underline{\text{TM}} \rightarrow \text{QC} ??$
Yes

- There does not exist a classical algorithm to efficiently simulate an arbitrary quantum system.
- Study problems that are BQP-hard (Bounded-error Quantum Polynomial).
- There exist problems that are even harder for quantum computers (QMA-hard).

"Post-Quantum Crypt"